

Yuri Pirola / Curriculum Vitae

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Research Interests

Algorithms and data structures in Bioinformatics / Computational and parameterized complexity / Genomic and transcriptomic sequence analysis and assembly / Haplotype inference and reconstruction / Evolutionary metaheuristics and genetic programming

Positions

- > Associate Professor / Oct. 2022–current
Dept. of Informatics, Systems and Communications, Univ. degli Studi di Milano–Bicocca
Member of the Bioinformatics and Experimental Algorithmic lab and of the Interdisciplinary Research Centre “Bicocca Bioinformatics Biostatistics and Bioimaging centre” (B4) at Univ. degli Studi di Milano–Bicocca
- > Assistant Professor / Oct. 2019–Sept. 2022
Dept. of Informatics, Systems and Communications, Univ. degli Studi di Milano–Bicocca
- > Post-Doctoral Fellow / May 2012–Dec. 2015 / “*Algorithmic methods for Next-Generation Sequencing data analysis*”
Funding: Univ. degli Studi di Milano–Bicocca
- > Post-Doctoral Fellow / Jan. 2011–Apr. 2012 / “*Efficient haplotype inference in livestock from high-density SNP chips*”
Funding: Univ. degli Studi di Milano–Bicocca / Lombardy Region / Parco Tecnologico Padano (Lodi)
- > Post-Doctoral Researcher / Mar. 2010–Dec. 2010 / “*PROZOO Project*”
Funding: Parco Tecnologico Padano (Lodi)

Research projects

- > “*FORGENOM II: Fostering Excellence in Advanced Genomics and Proteomics Research*”
Funding: EU Horizon WIDERA 2023 (2024–2027) / Role: WP leader
- > “*PINC: Pangenome INformatiCs from Theory to Applications*”
Funding: MUR (Italian Ministry of University), PRIN 2022 PNRR Action (2024–2025) / Role: participant
- > “*PANGAIA: Pan-genome Graph Algorithms and Data Integration*”
Funding: EU Horizon 2020, RISE Marie Skłodowska–Curie Action (2020–2025) / Role: participant
- > “*CORSAI: Raman analysis of saliva from COPD patients as new biomarker*”
Funding: ERA PerMed / Role: participant
- > “*Modulation of anti-cancer immune response by regulatory non-coding RNAs*”
Funding: Cariplo Foundation / Role: participant
- > “*Automata and formal languages: mathematical and applicative aspects*”
Funding: MIUR (Italian Ministry of Education), PRIN 2010/11 Action / Role: participant
- > “*Next Generation methods to preserve farm animal biodiversity by optimizing present and future breeding options*”
Funding: European Commission, 7th Framework Programme / Role: participant
- > “*PROZOO: Applications of genomics to fertility, disease resistance, and product quality assurance in cattle and pigs*”
Funding: Cariplo Foundation & Lombardy Region / Role: participant

Education

- > Ph.D. in Computer Science / Univ. degli Studi di Milano-Bicocca / February 3, 2010
“Combinatorial Problems in Studies of Genetic Variations: Haplotyping and Transcript Analysis”
 Supervisor: Prof. Paola Bonizzoni
- > M.Sc. in Computer Science, magna cum laude / Univ. degli Studi di Milano-Bicocca / February 23, 2006
“Analisi della Neutralità degli Spazi di Ricerca Booleani in Programmazione Genetica”
 Supervisors: Dr. Leonardo Vanneschi & Prof. Giancarlo Mauri

Fellowships

- > Ph.D. fellowship / MIUR (Italian Ministry of Education) / Nov. 2006–Oct. 2009.
- > Scholarship for graduate students / Univ. degli Studi di Milano / May–Oct. 2006

Publications

International peer-reviewed journal articles

- [J32] Ciccolella, S., Denti, L., Avila Cartes, J., Della Vedova, G., **Pirola, Y.**, Rizzi, R., and Bonizzoni, P. “Differential analysis of alternative splicing events in gene regions using residual neural networks”. *Neural Comput. Applic.* 37 (2025), 6819–6829. DOI: [10.1007/s00521-025-10992-2](https://doi.org/10.1007/s00521-025-10992-2). 
- [J31] Avila Cartes, J., Bonizzoni, P., Ciccolella, S., Della Vedova, G., Denti, L., Didelot, X., Monti, D. C., and **Pirola, Y.** “RecGraph: recombination-aware alignment of sequences to variation graphs”. *Bioinformatics* 40.5 (2024), btae292. DOI: [10.1093/bioinformatics/btae292](https://doi.org/10.1093/bioinformatics/btae292). 
- [J30] Bonizzoni, P., Costantini, M., De Felice, C., Petescia, A., **Pirola, Y.**, Previtali, M., Rizzi, R., Stoye, J., Zaccagnino, R., and Zizza, R. “Numeric Lyndon-based feature embedding of sequencing reads for machine learning approaches”. *Inf. Sci.* 607 (2022), 458–476. DOI: [10.1016/j.ins.2022.06.005](https://doi.org/10.1016/j.ins.2022.06.005). arXiv: 2202.13884 [q-bio.GN]. 
- [J29] Ciccolella, S., Denti, L., Bonizzoni, P., Della Vedova, G., **Pirola, Y.**, and Previtali, M. “MALVIRUS: an integrated application for viral variant analysis”. *BMC Bioinformatics* 22.15 (2022), 625. DOI: [10.1186/s12859-022-04668-0](https://doi.org/10.1186/s12859-022-04668-0). 
- [J28] Baaijens, J. A., Bonizzoni, P., Boucher, C., Della Vedova, G., **Pirola, Y.**, Rizzi, R., and Sirén, J. “Computational graph pangenomics: a tutorial on data structures and their applications”. *Nat. Comput.* 21 (2022), 81–108. DOI: [10.1007/s11047-022-09882-6](https://doi.org/10.1007/s11047-022-09882-6). 
- [J27] Bonizzoni, P., Della Vedova, G., **Pirola, Y.**, Previtali, M., and Rizzi, R. “Computing the multi-string BWT and LCP array in external memory”. *Theor. Comput. Sci.* 862 (2021), 42–58. DOI: [10.1016/j.tcs.2020.11.041](https://doi.org/10.1016/j.tcs.2020.11.041).
- [J26] Denti, L., **Pirola, Y.**, Previtali, M., Ceccato, T., Della Vedova, G., Rizzi, R., and Bonizzoni, P. “Shark: fishing relevant reads in an RNA-Seq sample”. *Bioinformatics* 37.4 (2021), 464–472. DOI: [10.1093/bioinformatics/btaa779](https://doi.org/10.1093/bioinformatics/btaa779). 
- [J25] Rizzi, R., Beretta, S., Patterson, M., **Pirola, Y.**, Previtali, M., Della Vedova, G., and Bonizzoni, P. “Overlap graphs and de Bruijn graphs: data structures for de novo genome assembly in the big data era”. *Quant. Biol.* 7.4 (2019), 278–292. DOI: [10.1007/s40484-019-0181-x](https://doi.org/10.1007/s40484-019-0181-x).
- [J24] Calabria, A., Beretta, S., Merelli, I., Spinozzi, G., Brasca, S., **Pirola, Y.**, Benedicenti, F., Tenderini, E., Bonizzoni, P., Milanesi, L., and Montini, E. “ γ -TRIS: a graph-algorithm for comprehensive identification of vector genomic insertion sites”. *Bioinformatics* 36.5 (2020), 1622–1624. DOI: [10.1093/bioinformatics/btz747](https://doi.org/10.1093/bioinformatics/btz747). 
- [J23] Bonizzoni, P., Della Vedova, G., **Pirola, Y.**, Previtali, M., and Rizzi, R. “Multithread Multistring Burrows–Wheeler Transform and Longest Common Prefix Array”. *J. Comput. Biol.* 26.9 (2019), 948–961. DOI: [10.1089/cmb.2018.0230](https://doi.org/10.1089/cmb.2018.0230).
- [J22] Grüning, B., Dale, R., Sjödin, A., Chapman, B. A., Rowe, J., Tomkins-Tinch, C. H., Valieris, R., Köster, J., and The Bioconda Team (including **Pirola, Y.**) “Bioconda: sustainable and comprehensive software distribution for the life sciences”. *Nature Methods* 15.7 (2018), 475–476. DOI: [10.1038/s41592-018-0046-7](https://doi.org/10.1038/s41592-018-0046-7).

- [J21] Bonizzoni, P., Della Vedova, G., **Pirola, Y.**, Previtali, M., and Rizzi, R. “FSG: Fast String Graph Construction for De Novo Assembly”. *J. Comput. Biol.* 24.10 (2017), 953–968. DOI: [10.1089/cmb.2017.0089](https://doi.org/10.1089/cmb.2017.0089).
- [J20] Bonizzoni, P., Della Vedova, G., **Pirola, Y.**, Previtali, M., and Rizzi, R. “An External-Memory Algorithm for String Graph Construction”. *Algorithmica* 78.2 (2017), 394–424. DOI: [10.1007/s00453-016-0165-4](https://doi.org/10.1007/s00453-016-0165-4).
- [J19] Biscarini, F., Schwarzenbacher, H., Pausch, H., Nicolazzi, E. L., **Pirola, Y.**, and Biffani, S. “Use of SNP genotypes to identify carriers of harmful recessive mutations in cattle populations”. *BMC Genomics* 17 (2016), 857. DOI: [10.1186/s12864-016-3218-9](https://doi.org/10.1186/s12864-016-3218-9). 
- [J18] Chiaradonna, F., **Pirola, Y.**, Ricciardiello, F., and Palorini, R. “Transcriptional profiling of immortalized and K-ras-transformed mouse fibroblasts upon PKA stimulation by forskolin in low glucose availability”. *Genomics Data* 9 (2016), 100–104. DOI: [10.1016/j.gdata.2016.07.004](https://doi.org/10.1016/j.gdata.2016.07.004). 
- [J17] Bonizzoni, P., Dondi, R., Klau, G. W., **Pirola, Y.**, Pisanti, N., and Zaccaria, S. “On the Minimum Error Correction Problem for Haplotype Assembly in Diploid and Polyploid Genomes”. *J. Comput. Biol.* 23.9 (2016), 718–736. DOI: [10.1089/cmb.2015.0220](https://doi.org/10.1089/cmb.2015.0220).
- [J16] Palorini, R., Votta, G., **Pirola, Y.**, De Vitto, H., De Palma, S., Airoidi, C., Vasso, M., Ricciardiello, F., Lombardi, P. P., Cirulli, C., Rizzi, R., Nicotra, F., Hiller, K., Gelfi, C., Alberghina, L., and Chiaradonna, F. “Protein Kinase A Activation Promotes Cancer Cell Resistance to Glucose Starvation and *Anoikis*”. *PLoS Genet.* 12.3 (2016), 1–41. DOI: [10.1371/journal.pgen.1005931](https://doi.org/10.1371/journal.pgen.1005931). 
- [J15] Bonizzoni, P., Della Vedova, G., **Pirola, Y.**, Previtali, M., and Rizzi, R. “LSG: An External-Memory Tool to Compute String Graphs for NGS Data Assembly”. *J. Comput. Biol.* 23.3 (2016), 137–149. DOI: [10.1089/cmb.2015.0172](https://doi.org/10.1089/cmb.2015.0172).
- [J14] **Pirola, Y.**, Zaccaria, S., Dondi, R., Klau, G. W., Pisanti, N., and Bonizzoni, P. “HapCol: Accurate and Memory-Efficient Haplotype Assembly from Long Reads”. *Bioinformatics* 32.11 (2016), 1610–1617. DOI: [10.1093/bioinformatics/btv495](https://doi.org/10.1093/bioinformatics/btv495). 
- [J13] Beerenwinkel, N., Beretta, S., Bonizzoni, P., Dondi, R., and **Pirola, Y.** “Covering Pairs in Directed Acyclic Graphs”. *Comput. J.* 58.7 (2015), 1673–1686. DOI: [10.1093/comjnl/bxu116](https://doi.org/10.1093/comjnl/bxu116).
- [J12] Batini, C., Bonizzoni, P., Comerio, M., Dondi, R., **Pirola, Y.**, and Salandra, F. “A Clustering Algorithm for Planning the Integration Process of a Large Number of Conceptual Schemas”. *J. Comput. Sci. Technol.* 30.1 (2015), 214–224. DOI: [10.1007/s11390-015-1514-5](https://doi.org/10.1007/s11390-015-1514-5).
- [J11] Beretta, S., Bonizzoni, P., Della Vedova, G., **Pirola, Y.**, and Rizzi, R. “Modeling Alternative Splicing Variants from RNA-Seq Data with Isoform Graphs”. *J. Comput. Biol.* 21.1 (2014), 16–40. DOI: [10.1089/cmb.2013.0112](https://doi.org/10.1089/cmb.2013.0112). 
- [J10] Bonizzoni, P., Della Vedova, G., Dondi, R., and **Pirola, Y.** “Parameterized Complexity of k -Anonymity: Hardness and Tractability”. *J. Comb. Optim.* 26.1 (2013), 19–43. DOI: [10.1007/s10878-011-9428-9](https://doi.org/10.1007/s10878-011-9428-9).
- [J9] Bonizzoni, P., Dondi, R., and **Pirola, Y.** “Maximum Disjoint Paths on Edge-Colored Graphs: Approximability and Tractability”. *Algorithms* 6.1 (2013), 1–11. DOI: [10.3390/a6010001](https://doi.org/10.3390/a6010001). 
- [J8] **Pirola, Y.**, Della Vedova, G., Biffani, S., Stella, A., and Bonizzoni, P. “A Fast and Practical Approach to Genotype Phasing and Imputation on a Pedigree with Erroneous and Incomplete Information”. *IEEE/ACM Trans. Comput. Biol. Bioinform.* 9.6 (2012), 1582–1594. DOI: [10.1109/TCBB.2012.100](https://doi.org/10.1109/TCBB.2012.100).
- [J7] **Pirola, Y.**, Rizzi, R., Picardi, E., Pesole, G., Della Vedova, G., and Bonizzoni, P. “PItron: A Fast Method for Detecting the Gene Structure Due to Alternative Splicing Via Maximal Pairings of a Pattern and a Text”. *BMC Bioinformatics* 13.S5 (2012), S2. DOI: [10.1186/1471-2105-13-S5-S2](https://doi.org/10.1186/1471-2105-13-S5-S2). 
- [J6] Vanneschi, L., **Pirola, Y.**, Mauri, G., Tomassini, M., Collard, P., and Verel, S. “A Study of Neutrality of Boolean Function Landscapes in Genetic Programming”. *Theor. Comput. Sci.* 425 (2012), 34–57. DOI: [10.1016/j.tcs.2011.03.011](https://doi.org/10.1016/j.tcs.2011.03.011). 
- [J5] **Pirola, Y.**, Bonizzoni, P., and Jiang, T. “An Efficient Algorithm for Haplotype Inference on Pedigrees with Recombinations and Mutations”. *IEEE/ACM Trans. Comput. Biol. Bioinform.* 9.1 (2012), 12–25. DOI: [10.1109/TCBB.2011.51](https://doi.org/10.1109/TCBB.2011.51).

- [J4] Bonizzoni, P., Della Vedova, G., Dondi, R., and Pirola, Y. “Variants of Constrained Longest Common Subsequence”. *Inf. Process. Lett.* 110.20 (2010), 877–881. DOI: [10.1016/j.ipl.2010.07.015](https://doi.org/10.1016/j.ipl.2010.07.015).
- [J3] Bonizzoni, P., Della Vedova, G., Dondi, R., Pirola, Y., and Rizzi, R. “Pure Parsimony Xor Haplotyping”. *IEEE/ACM Trans. Comput. Biol. Bioinform.* 7.4 (2010), 598–610. DOI: [10.1109/TCBB.2010.52](https://doi.org/10.1109/TCBB.2010.52).
- [J2] Della Vedova, G., Dondi, R., Jiang, T., Pavesi, G., Pirola, Y., and Wang, L. “Beyond Evolutionary Trees”. *Nat. Comput.* 9.2 (2010), 421–435. DOI: [10.1007/s11047-009-9156-6](https://doi.org/10.1007/s11047-009-9156-6).
- [J1] Bonizzoni, P., Mauri, G., Pesole, G., Picardi, E., Pirola, Y., and Rizzi, R. “Detecting Alternative Gene Structures from Spliced ESTs: A Computational Approach”. *J. Comput. Biol.* 16.1 (2009), 43–66. DOI: [10.1089/cmb.2008.0028](https://doi.org/10.1089/cmb.2008.0028).

International peer-reviewed conference papers

- [C20] Bonizzoni, P., De Felice, C., Pirola, Y., Rizzi, R., Zaccagnino, R., and Zizza, R. “Identification of Chimeric RNAs: A Novel Machine Learning Perspective”. In: *Computational Advances in Bio and medical Sciences (IC-CABS), Revised selected papers*. Vol. 14548. LNCS. Springer, 2025, 14–26. DOI: [10.1007/978-3-031-82768-6_2](https://doi.org/10.1007/978-3-031-82768-6_2).
- [C19] Bonizzoni, P., Boucher, C., Cozzi, D., Gagic, T., and Pirola, Y. “Solving the Minimal Positional Substring Cover Problem in Sublinear Space”. In: *Combinatorial Pattern Matching (CPM)*. Vol. 296. LIPIcs. Schloss Dagstuhl – LZI, 2024, 12:1–12:16. DOI: [10.4230/LIPICS.CPM.2024.12](https://doi.org/10.4230/LIPICS.CPM.2024.12). 
- [C18] Bonizzoni, P., Della Vedova, G., Pirola, Y., Rizzi, R., and Sgrò, M. “Multiallelic Maximal Perfect Haplotype Blocks with Wildcards via PBWT”. In: *Bioinformatics and Biomedical Engineering (IWBBIO)*. Vol. 13919. LNCS. Springer, 2023, 3–12. DOI: [10.1007/978-3-031-34953-9_5](https://doi.org/10.1007/978-3-031-34953-9_5).
- [C17] Bonizzoni, P., Petescia, A., Pirola, Y., Rizzi, R., Zaccagnino, R., and Zizza, R. “KFinger: Capturing Overlaps Between Long Reads by Using Lyndon Fingerprints”. In: *Bioinformatics and Biomedical Engineering (IWBBIO)*. Vol. 13347. LNCS. Springer, 2022, 3–12. DOI: [10.1007/978-3-031-07802-6_37](https://doi.org/10.1007/978-3-031-07802-6_37).
- [C16] Bonizzoni, P., De Felice, C., Pirola, Y., Rizzi, R., Zaccagnino, R., and Zizza, R. “Can Formal Languages Help Pangenomics to Represent and Analyze Multiple Genomes?” In: *Developments in Language Theory (DLT)*. Vol. 13257. LNCS. Springer, 2022, 3–12. DOI: [10.1007/978-3-031-05578-2_1](https://doi.org/10.1007/978-3-031-05578-2_1).
- [C15] Bonizzoni, P., De Felice, C., Petescia, A., Pirola, Y., Rizzi, R., Stoye, J., Zaccagnino, R., and Zizza, R. “Can We Replace Reads by Numeric Signatures? Lyndon Fingerprints as Representations of Sequencing Reads for Machine Learning”. In: *Algorithms for Computational Biology (ALCoB)*. Vol. 12715. LNCS. Springer, 2021, 16–28. DOI: [10.1007/978-3-030-74432-8_2](https://doi.org/10.1007/978-3-030-74432-8_2).
- [C14] Bonizzoni, P., Della Vedova, G., Nicosia, S., Pirola, Y., Previtali, M., and Rizzi, R. “Divide and Conquer Computation of the Multi-string BWT and LCP Array”. In: *Computability in Europe (CiE)*. Vol. 10936. LNCS. Springer, 2018, 107–117. DOI: [10.1007/978-3-319-94418-0_11](https://doi.org/10.1007/978-3-319-94418-0_11).
- [C13] Bonizzoni, P., Della Vedova, G., Pirola, Y., Previtali, M., and Rizzi, R. “FSG: Fast String Graph Construction for De Novo Assembly of reads data”. In: *Bioinformatics Research and Applications (ISBRA)*. Vol. 9683. LNCS. Springer, 2016, 27–39. DOI: [10.1007/978-3-319-38782-6_3](https://doi.org/10.1007/978-3-319-38782-6_3).
- [C12] Bonizzoni, P., Dondi, R., Klau, G. W., Pirola, Y., Pisanti, N., and Zaccaria, S. “On the Fixed Parameter Tractability and Approximability of the Minimum Error Correction problem”. In: *Combinatorial Pattern Matching (CPM)*. Vol. 9133. LNCS. Springer, 2015, 100–113. DOI: [10.1007/978-3-319-19929-0_9](https://doi.org/10.1007/978-3-319-19929-0_9).
- [C11] Bonizzoni, P., Della Vedova, G., Pirola, Y., Previtali, M., and Rizzi, R. “Constructing String Graphs in External Memory”. In: *Algorithms in Bioinformatics (WABI)*. Vol. 8701. LNCS. Springer, 2014, 311–325. DOI: [10.1007/978-3-662-44753-6_23](https://doi.org/10.1007/978-3-662-44753-6_23).
- [C10] Beerenwinkel, N., Beretta, S., Bonizzoni, P., Dondi, R., and Pirola, Y. “Covering Pairs in Directed Acyclic Graphs”. In: *Language and Automata Theory and Applications (LATA)*. Vol. 8370. LNCS. Springer, 2014, 126–137. DOI: [10.1007/978-3-319-04921-2_10](https://doi.org/10.1007/978-3-319-04921-2_10).
- [C9] Pirola, Y., Della Vedova, G., Bonizzoni, P., Stella, A., and Biscarini, F. “Haplotype-based prediction of gene alleles using pedigrees and SNP genotypes”. In: *Bioinformatics, Computational Biology, and Biomedical Informatics (ACM BCB)*. ACM, 2013, 33–41. DOI: [10.1145/2506583.2506592](https://doi.org/10.1145/2506583.2506592).

- [C8] Pirola, Y., Della Vedova, G., Biffani, S., Stella, A., and Bonizzoni, P. “A fast and practical approach to genotype phasing and imputation on a pedigree with erroneous and incomplete information”. In: *Computational Advances in Bio and medical Sciences (ICCABS)*. IEEE, 2012. DOI: [10.1109/ICCABS.2012.6182643](https://doi.org/10.1109/ICCABS.2012.6182643).
- [C7] Bonizzoni, P., Della Vedova, G., Pirola, Y., and Rizzi, R. “PItron: a fast method for gene structure prediction via maximal pairings of a pattern and a text”. In: *Computational Advances in Bio and medical Sciences (ICCABS)*. IEEE, 2011, 33–39. DOI: [10.1109/ICCABS.2011.5729935](https://doi.org/10.1109/ICCABS.2011.5729935).
- [C6] Pirola, Y., Bonizzoni, P., and Jiang, T. “Haplotype Inference on Pedigrees with Recombinations and Mutations”. In: *Algorithms in Bioinformatics (WABI)*. Vol. 6293. LNCS. Springer, 2010, 148–161. DOI: [10.1007/978-3-642-15294-8_13](https://doi.org/10.1007/978-3-642-15294-8_13).
- [C5] Bonizzoni, P., Della Vedova, G., Dondi, R., and Pirola, Y. “Parameterized Complexity of k-Anonymity: Hardness and Tractability”. In: *Combinatorial Algorithms (IWOCA)*. Vol. 6460. LNCS. Springer, 2011, 242–255. DOI: [10.1007/978-3-642-19222-7_25](https://doi.org/10.1007/978-3-642-19222-7_25).
- [C4] Bonizzoni, P., Della Vedova, G., Dondi, R., Pirola, Y., and Rizzi, R. “Minimum Factorization Agreement of Spliced ESTs”. In: *Algorithms in Bioinformatics (WABI)*. Vol. 5724. LNCS. Springer, 2009, 1–12. DOI: [10.1007/978-3-642-04241-6_1](https://doi.org/10.1007/978-3-642-04241-6_1).
- [C3] Bonizzoni, P., Della Vedova, G., Dondi, R., Pirola, Y., and Rizzi, R. “Pure Parsimony Xor Haplotyping”. In: *Bioinformatics Research and Applications (ISBRA)*. Vol. 5542. LNCS. Springer, 2009, 186–197. DOI: [10.1007/978-3-642-01551-9_19](https://doi.org/10.1007/978-3-642-01551-9_19).
- [C2] Vanneschi, L., Tomassini, M., Collard, P., Verel, S., Pirola, Y., and Mauri, G. “A Comprehensive View of Fitness Landscapes with Neutrality and Fitness Clouds”. In: *Genetic Programming (EuroGP)*. Vol. 4445. LNCS. Springer, 2007, 241–250. DOI: [10.1007/978-3-540-71605-1_22](https://doi.org/10.1007/978-3-540-71605-1_22).
- [C1] Vanneschi, L., Pirola, Y., and Collard, P. “A quantitative study of neutrality in GP boolean landscapes”. In: *Genetic and Evolutionary Computation (GECCO)*. ACM, 2006, 895–902. DOI: [10.1145/1143997.1144152](https://doi.org/10.1145/1143997.1144152).

Book chapters

- [B2] Dondi, R. and Pirola, Y. “Beyond Evolutionary Trees”. In: *Encyclopedia of Algorithms*. Ed. by M.-Y. Kao. Springer, 2016, 183–189. ISBN: 978-3-642-27848-8. DOI: [10.1007/978-3-642-27848-8_599-1](https://doi.org/10.1007/978-3-642-27848-8_599-1).
- [B1] Bonizzoni, P., Della Vedova, G., Pesole, G., Picardi, E., Pirola, Y., and Rizzi, R. “Transcriptome Assembly and Alternative Splicing Analysis”. In: *RNA Bioinformatics*. Ed. by E. Picardi. Vol. 1269. Methods in Molecular Biology. Springer, 2015, 173–188. ISBN: 978-1-4939-2290-1. DOI: [10.1007/978-1-4939-2291-8_11](https://doi.org/10.1007/978-1-4939-2291-8_11).

Talks, schools, and visits

Invited talks

- > Università degli Studi di Milano / invited by Prof. Giovanni Righini / Feb. 8, 2010
- > Parco Tecnologico Padano (Lodi) / invited by Dr. Alessandra Stella / Mar. 25, 2010

Talks

- > *Bioinformatics and Biomedical Engineering (IWBBIO)* / Gran Canaria, Spain / Jun. 2022
- > *Bioinformatics and Computational Biology Conference (BBCC)* / Naples, Italy / Nov. 2020
- > *Bioinformatics Open Source Conference (BOSC)* / Toronto, Canada / Jul. 2020
- > *Int. Workshop on Data Structures in Bioinformatics (DSB)* / Rennes, France / Feb. 2020
- > *Int. Conf. on Language and Automata Theory and Applications (LATA)* / Madrid, Spain / Mar. 2014
- > *Workshop “Combinatorial structures for sequence analysis in bioinformatics”* / Milan, Italy / Nov. 2013

- › *ACM Int. Conf. on Bioinf., Computational Biol., and Biomedical Inform. (ACM BCB)* / Washington DC, USA / Sep. 2013
- › *Italian Conf. on Theoretical Computer Science (ICTCS)* / Palermo, Italy / Sep. 2013
- › *IEEE Int. Conf. on Computational Advances in Bio and medical Sciences (ICCABS)* / Las Vegas NV, USA / Feb. 2012
- › *IEEE Int. Conf. on Computational Advances in Bio and medical Sciences (ICCABS)* / Orlando FL, USA / Feb. 2011
- › *Int. Workshop on Algorithms in Bioinformatics (WABI)* / Liverpool, UK / Sep. 2010
- › *Int. Workshop on Algorithms in Bioinformatics (WABI)* / Philadelphia PA, USA / Sep. 2009
- › *Int. Symp. on Bioinformatics Research and Applications (ISBRA)* / Ft. Lauderdale FL, USA / May 2009

Schools

- › *Lipari Int. Summer School on Bioinformatics and Computational Biology* / Univ. degli Studi di Catania / 2008
- › *Summer school on Parallel and Scientific Computing* / CINECA / 2006

Research visits

- › *Centrum Wiskunde & Informatica* / Amsterdam, The Netherlands / Dr. Gunnar W. Klau / Jul. 14–17, 2015
- › *Université Paris-Est Marne-la-Vallée* / Paris, France / Dr. Gregory Kucherov / Jan. 12–18, 2014
- › *University of California, Riverside* / Riverside CA, USA / Prof. Tao Jiang / Feb.–Jun. 2009

Professional activities

- › Local Technical Coordinator (LTec) for Univ. of Milano-Bicocca of the Italian node of ELIXIR
- › Co-organizer (with S. Pissis, CWI) of a satellite workshop of ECCB 2020
- › *Program committee* member of: WABI 2024–2025 / BICOB 2020–2026 / IWBBIO 2020, 2022–2026 / BBC 2015–2025 (part of ICCS) / ICTCS 2022 / HPC4COVID-19 2020 (part of IEEE BIBM) / PDP 2015–2017
- › *Organizing committee* member of: CPM 2025 / PhD School “Introduction to Pangenomics” 2022 / DSB 2021 / CiE 2013 / “Unconventional Models of Computation” 2009 / ASWorkshop 2008
- › Member of the reviewer board of Algorithms
- › *Reviewer* for the following journals: Discrete Applied Mathematics / Bioinformatics / Briefings in Bioinformatics / IEEE-ACM Trans. Computational Biology and Bioinformatics / IEEE Access / J. Computational Science / J. Biomedical Informatics / Computers & Operations Research / Frontiers in Bioinformatics / Int. Trans. in Operational Research / GigaScience / Applied Sciences / Mathematics / Genes / Processes / Systems / Technologies / Concurrency and Computation: Practice and Experience / BMC Genomics / BMC Genomic Data / Int. J. Bioinformatics Research and Applications / Int. J. Molecular Sciences / BioMed Research Int.
- › *Reviewer* for the following international conferences: IWOCA 2025 / SPIRE 2023–2024 / RECOMB 2023 / IEEE BIBM 2022, 2021, 2020, 2013, 2012, 2009 / WABI 2022, 2021, 2014, 2012 / ISBRA 2021, 2015, 2014 / RECOMB-CG 2021 / APBC 2020 / CIBB 2019, 2014 / SODA 2018 / IWOCA 2016, 2015 / AAIM 2016 / WALCOM 2016 / BSB 2013 / CiE 2013 / CATS 2013 / ECCB 2012 / ICCABS 2011

Supervision activities

- › Co-supervisor for 1 Ph.D. thesis in CS at Univ. degli Studi di Milano-Bicocca
- › Supervisor for 6 M.Sc. theses in CS at Univ. degli Studi di Milano-Bicocca
- › Supervisor for 9 B.Sc. theses in CS at Univ. degli Studi di Milano-Bicocca

Teaching

- > Member of the board of the Ph.D. program in Computer Science at Univ. degli Studi di Milano-Bicocca
- > *Teacher/Teaching assistant*, at Univ. degli Studi di Milano-Bicocca, of:

Theory of computing	8h, class	M.Sc.	School of Computer Science	2025/26
Algorithms and DS	52h, class	B.Sc.	School of Computer Science	2025/26
Algorithms and DS	48h, lab.	B.Sc.	School of Computer Science	2025/26
Software design	12h, lab.	M.Sc.	School of Computer Science	2025/26
Theory of computing	16h, class	M.Sc.	School of Computer Science	2022/23–24/25
Algorithms and DS	52h, class	B.Sc.	School of Computer Science	2022/23–24/25
Algorithms and DS	40h, lab.	B.Sc.	School of Computer Science	2022/23–24/25
Software design	12h, lab.	M.Sc.	School of Computer Science	2019/20–24/25
Computer science	16h, class	MD	School of Medicine	2018/19–21/22
Genome assembly	20h, class	2nd level master (EQF/RQF level 8)		2020/21
Algorithms and DS	20h, class	B.Sc.	School of Computer Science	2019/20–21/22
Algorithms and DS	20h, lab.	B.Sc.	School of Computer Science	2019/20–21/22
Elements of Bioinformatics	16h, lab.	B.Sc.	School of Computer Science	2013/14
Algorithms and DS 2	12h, class	B.Sc.	School of Computer Science	2010/11
Bioinformatics	4h, lectures	M.Sc.	School of Computer Science	2009/10
Bioinformatics	8h, class	M.Sc.	School of Computer Science	2008/09
Analysis of algorithms	12h, class	M.Sc.	School of Computer Science	2007/08
Bioinformatics	4h, lectures	B.Sc.	School of Computer Science	2007/08
Programming languages	24h, lab.	B.Sc.	School of Computer Science	2006/07
Algorithms and DS	12h, lab.	B.Sc.	School of Computer Science	2005/06
Operating systems	24h, lab.	B.Sc.	School of Computer Science	2005/06
Operating systems	48h, lab.	B.Sc.	School of Computer Science	2004/05
- > *Teaching assistant*, at Univ. degli Studi di Bergamo, of:

Introduction to Informatics	30h, class	B.A.	School of Foreign Languages	2007/08–08/09
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- > *Lecturer* for the course “Population-based Optimisation Methods” (4h, June 2022) of the Ph.D. program in Computer Science at Univ. degli Studi di Milano-Bicocca
- > *Lecturer* for the course “Do we need data structures?” (4h, Oct. 2020) of the Ph.D. program in Computer Science at Univ. degli Studi di Milano-Bicocca
- > *Lecturer* on the topic “Introduction to programming in C” (16h) for post-graduate LS researchers at PTP, Lodi